

- AES-128->AES-256

AES-128->SPECK

ChaCha20->SPECK

SPECK->HIGHT

AES-128->HIGHT

ChaCha20->AES-256

HIGHT->RECTANGLE
- ### Encryption

This line graph illustrates the memory overhead (in KB, log scale) for various encryption combinations as data size increases from 1KB to 50MB. The y-axis is logarithmic, ranging from 1 to over 1000 KB. The x-axis represents data sizes: 1KB, 5KB, 10KB, 50KB, 100KB, 1MB, 5MB, 10MB, and 50MB. All combinations show a sharp increase in memory overhead starting around 50KB, converging to approximately 4,745 KB at 50MB. The combinations are: AES-128->AES-256 (red solid line with circles), AES-128->SPECK (orange dashed line with triangles), ChaCha20->SPECK (yellow solid line with inverted triangles), AES-128->HIGHT (red dashed line with squares), ChaCha20->AES-256 (yellow dashed line with diamonds), and HIGHT->RECTANGLE (green dash-dot line with pluses). The SPECK->HIGHT combination (teal dash-dot line with crosses) is not plotted in this graph.
- ### Decryption

This line graph illustrates the memory overhead (in KB, log scale) for various decryption combinations as data size increases from 1KB to 50MB. The y-axis is logarithmic, ranging from 1 to over 1000 KB. The x-axis represents data sizes: 1KB, 5KB, 10KB, 50KB, 100KB, 1MB, 5MB, 10MB, and 50MB. All combinations show a sharp increase in memory overhead starting around 50KB, converging to approximately 4,745 KB at 50MB. The combinations are: AES-128->AES-256 (red solid line with circles), AES-128->SPECK (orange dashed line with triangles), ChaCha20->SPECK (yellow solid line with inverted triangles), AES-128->HIGHT (red dashed line with squares), ChaCha20->AES-256 (yellow dashed line with diamonds), and HIGHT->RECTANGLE (green dash-dot line with pluses). The SPECK->HIGHT combination (teal dash-dot line with crosses) is not plotted in this graph.
- | | 1KB | 5KB | 10KB | 50KB | 100KB | 1MB | 5MB | 10MB | 50MB |
|-------------------|-------|------|------|------|-------|-------|-----|------|-------|
| AES-128->AES-256 | 2.72 | 2.90 | 3.14 | 5.01 | 7.36 | 97.54 | 477 | 951 | 4,746 |
| AES-128->HIGHT | 1.79 | 1.98 | 2.22 | 4.09 | 6.44 | 96.62 | 476 | 950 | 4,745 |
| AES-128->SPECK | 1.88 | 2.07 | 2.30 | 4.18 | 6.52 | 96.71 | 476 | 951 | 4,746 |
| ChaCha20->AES-256 | 2.72 | 2.91 | 3.14 | 5.02 | 7.36 | 97.55 | 477 | 951 | 4,746 |
| ChaCha20->SPECK | 1.88 | 2.07 | 2.31 | 4.18 | 6.52 | 96.71 | 476 | 951 | 4,746 |
| HIGHT->RECTANGLE | 0.912 | 1.10 | 1.33 | 3.21 | 5.55 | 95.74 | 475 | 950 | 4,745 |
| SPECK->HIGHT | 0.976 | 1.12 | 1.35 | 3.22 | 5.57 | 95.76 | 475 | 950 | 4,745 |
- | | 1KB | 5KB | 10KB | 50KB | 100KB | 1MB | 5MB | 10MB | 50MB |
|-------------------|-------|------|------|------|-------|-------|-----|------|-------|
| AES-128->AES-256 | 2.18 | 2.36 | 2.59 | 4.47 | 6.81 | 97.00 | 476 | 951 | 4,746 |
| AES-128->HIGHT | 1.48 | 1.67 | 1.90 | 3.78 | 6.12 | 96.31 | 476 | 950 | 4,745 |
| AES-128->SPECK | 1.59 | 1.78 | 2.01 | 3.89 | 6.23 | 96.42 | 476 | 950 | 4,745 |
| ChaCha20->AES-256 | 2.21 | 2.39 | 2.62 | 4.50 | 6.84 | 97.03 | 477 | 951 | 4,746 |
| ChaCha20->SPECK | 1.62 | 1.81 | 2.04 | 3.92 | 6.26 | 96.45 | 476 | 950 | 4,745 |
| HIGHT->RECTANGLE | 0.922 | 1.11 | 1.34 | 3.22 | 5.56 | 95.75 | 475 | 950 | 4,745 |
| SPECK->HIGHT | 0.960 | 1.12 | 1.35 | 3.23 | 5.57 | 95.76 | 475 | 950 | 4,745 |